Day 15 Study Notes

Fri 27 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Wave optics (part 1: interference) (75 min) · Maths: Statistics (75 min) · Chemistry: p-block elements (groups 13 to 18) (60 min)
Exam	1 h 15 min	The Day 15 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Wave optics (part 1: interference)

Wave optics gives one question in most papers, very often on Young's double-slit experiment. Today covers Huygens' principle and interference. Part 2 covers diffraction and polarisation.

1. Huygens' principle

- Every point on a wavefront acts as a source of secondary wavelets; the new wavefront is the surface touching all the wavelets.
- A point source gives spherical wavefronts, a line source cylindrical ones, and a very distant source plane wavefronts.
- Applied to refraction it gives Snell's law with $\dfrac{\sin i}{\sin r} = \dfrac{v_1}{v_2}$: light travels slower in the denser medium.

2. Superposition and coherence

- Two sources are **coherent** if they have the same frequency and a constant phase difference. Two separate lamps are never coherent; coherent sources are made by splitting one source (as with two slits).
- Resultant intensity: $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$.
- Constructive interference: $\phi = 2n\pi$, path difference $n\lambda$. $I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$.
- Destructive interference: $\phi = (2n+1)\pi$, path difference $(2n+1)\dfrac{\lambda}{2}$. $I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$.
- $\dfrac{I_{max}}{I_{min}} = \left(\dfrac{a_1 + a_2}{a_1 - a_2}\right)^2$, where amplitudes are in the ratio $\sqrt{I_1} : \sqrt{I_2}$.
- Phase difference $= \dfrac{2\pi}{\lambda} \times$ path difference.
- For two equal sources I_0: $I = 4I_0\cos^2\dfrac{\phi}{2}$.

3. Young's double-slit experiment

Slit separation d, screen distance D, wavelength λ.

- Path difference at a point y from the centre: $\dfrac{yd}{D}$.
- Bright fringes: $y_n = \dfrac{n\lambda D}{d}$. Dark fringes: $y_n = \dfrac{(2n-1)\lambda D}{2d}$ for the nth dark fringe.
- Fringe width $\beta = \dfrac{\lambda D}{d}$, the same for bright and dark fringes. Angular fringe width $= \dfrac{\lambda}{d}$.
- In a medium of index n: λ and so β become $\dfrac{1}{n}$ of their values in air.
- With white light the central fringe is white and the others are coloured, with red farthest out in each order.
- **Thin sheet** of thickness t and index μ over one slit: the whole pattern shifts towards that slit by $\dfrac{(\mu-1)tD}{d}$. The fringe width does not change.

> **Solved example 1.** In Young's experiment $\lambda = 600$ nm, $d = 1$ mm and $D = 1$ m. Find the fringe width.
> $\beta = \dfrac{600 \times 10^{-9} \times 1}{10^{-3}} = 6 \times 10^{-4}$ m $= 0.6$ mm.

Solved example 2. Two sources have intensities in the ratio 9 : 1. Find $\dfrac{I_{max}}{I_{min}}$.

Amplitudes are $3:1$, so the ratio is $\dfrac{(3+1)^2}{(3-1)^2} = \dfrac{16}{4} = 4$.

Traps. Use amplitudes, not intensities, in the I_{max}/I_{min} formula. A transparent sheet shifts the pattern but leaves the fringe width unchanged. In water the fringes get closer, not farther. The nth dark fringe is at $(2n-1)\dfrac{\lambda D}{2d}$, not $(2n+1)$.

Mathematics · Statistics

Statistics gives one question in nearly every paper, usually a mean and variance calculation with a twist: a wrongly recorded value, an observation removed, or a change of scale.

1. Measures of central tendency

- Mean: $\bar{x} = \dfrac{\sum x_i}{n}$, or $\dfrac{\sum f_i x_i}{\sum f_i}$ for a frequency table.
- Median: the middle value after sorting. For even n, the mean of the two middle values.
- Grouped median: $l + \dfrac{\frac{N}{2} - C}{f} \times h$, where l is the lower limit of the median class, C the cumulative frequency before it, f its frequency and h its width.
- Mode: the most frequent value. Empirical relation for moderately skewed data: mode $= 3$ median $- 2$ mean.
- Mean of the first n natural numbers: $\dfrac{n+1}{2}$.

2. Mean deviation

- About a point a: $\dfrac{\sum |x_i - a|}{n}$. Usually taken about the mean or the median.
- The mean deviation is least when taken about the median.

3. Variance and standard deviation

- $\sigma^2 = \dfrac{\sum (x_i - \bar{x})^2}{n} = \dfrac{\sum x_i^2}{n} - \bar{x}^2$. Standard deviation $\sigma = \sqrt{\sigma^2}$.
- Variance of the first n natural numbers: $\dfrac{n^2 - 1}{12}$.
- **Change of origin and scale:** if $y = ax + b$, then $\bar{y} = a\bar{x} + b$, $\sigma_y^2 = a^2 \sigma_x^2$ and $\sigma_y = |a| \sigma_x$. Adding a constant does not change the variance.
- Coefficient of variation $= \dfrac{\sigma}{\bar{x}} \times 100$. The series with the smaller CV is more consistent.

4. Combined and corrected values

- Combined mean of two groups: $\dfrac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$.
- Combined variance: $\dfrac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$, with $d_i = \bar{x}_i - \bar{x}$.
- **Wrong or removed observations:** work with n, $\sum x$ and $\sum x^2$. From the mean and variance recover $\sum x = n\bar{x}$ and $\sum x^2 = n(\sigma^2 + \bar{x}^2)$; correct the sums; recompute.

> **Solved example 1.** The mean of 10 observations is 15. One value, 25, was wrongly read as 52. Find the correct mean.
> Correct sum $= 150 - 52 + 25 = 123$. Mean $= 12.3$.

Solved example 2. Five observations have mean 10 and variance 4. The observation 10 is removed. Find the new variance.

$\sum x = 50$, $\sum x^2 = 5(4 + 100) = 520$. Removing 10: $\sum x = 40$, $\sum x^2 = 420$, $n = 4$. Mean $= 10$, variance $= \dfrac{420}{4} - 100 = 5$.

Traps. Multiplying by a multiplies the variance by a^2, not a. Adding a constant changes the mean but not the variance or the standard deviation. Sort the data before finding the median. When an observation is removed, change n too.

Chemistry · p-block elements (groups 13 to 18)

The p-block gives two to three questions per paper, almost all from NCERT statements and trends. This chapter was originally planned for Day 21 and moved here. Read the NCERT p-block chapters alongside; this sheet is the list of trends that are asked.

1. Group 13 (B, Al, Ga, In, Tl)

- **Inert pair effect:** going down the group, the ns^2 pair is less willing to bond, so the +1 state becomes more stable. Tl^{+} is more stable than Tl^{3+}.
- Atomic radius: Ga is slightly smaller than Al, because of the poor shielding by its 3d electrons.
- **Boric acid**, $B(OH)_3$, is a weak monobasic acid. It is a Lewis acid: it accepts OH^{-} from water rather than giving away its own H^{+}.
- **Diborane**, B_2H_6: four terminal B–H bonds and two bridging B–H–B bonds, each a three-centre two-electron ("banana") bond.
- Lewis acid strength: $BF_3 < BCl_3 < BBr_3$, because back-bonding from F to B is strongest and makes BF_3 the least electron-deficient.

2. Group 14 (C, Si, Ge, Sn, Pb)

- Catenation: C ≫ Si > Ge ≈ Sn. Carbon also forms pπ–pπ multiple bonds, which the heavier members do not.
- The +2 state becomes more stable down the group: Pb^{2+} is more stable than Pb^{4+}, so PbO_2 is a strong oxidising agent, while Sn^{2+} is a reducing agent.
- CO is neutral and poisonous (binds haemoglobin); CO_2 is acidic. SiO_2 is acidic. SnO, SnO_2, PbO and PbO_2 are amphoteric.

3. Group 15 (N, P, As, Sb, Bi)

- N_2 has a very strong triple bond and is unreactive; phosphorus exists as P_4 tetrahedra.
- Hydrides: basicity and thermal stability $NH_3 > PH_3 > AsH_3 > SbH_3$. Bond angle falls from 107° (NH_3) to about 91° (SbH_3). Reducing character rises down the group.
- Basicity of phosphorus oxoacids equals the number of P–OH groups: H_3PO_4 3, H_3PO_3 2, H_3PO_2 1. The P–H bonds make H_3PO_3 and H_3PO_2 reducing agents.
- P_4O_{10} has 6 P–O–P bridges and 4 terminal P=O bonds.

4. Group 16 (O, S, Se, Te, Po)

- Boiling points of hydrides: H_2O is highest (hydrogen bonding); then $H_2S < H_2Se < H_2Te$.
- Acid strength rises down the group: $H_2O < H_2S < H_2Se < H_2Te$, because the H–X bond gets weaker. Thermal stability falls the same way.
- Oxygen shows mainly −2; sulphur shows up to +6 because it has d orbitals.
- Ozone is a bent molecule and a powerful oxidising agent.

5. Group 17 (F, Cl, Br, I)

- Fluorine is the most electronegative element and shows only −1.

- Electron gain enthalpy is most negative for **chlorine**, not fluorine: fluorine is so small that added electrons repel strongly.
- Bond energy: $Cl_2 > Br_2 > F_2 > I_2$. F_2 is weak because of lone-pair repulsion in a short bond.
- F_2 is the strongest oxidising agent.
- Acid strength of HX in water: HF < HCl < HBr < HI. HF is weak because the H–F bond is very strong and it hydrogen-bonds.
- Oxoacids of chlorine: acid strength $HClO < HClO_2 < HClO_3 < HClO_4$.

6. Group 18 (noble gases)

Compound	Bond pairs + lone pairs on Xe	Shape
XeF_2	2 + 3	linear
XeF_4	4 + 2	square planar
XeF_6	6 + 1	distorted octahedral
XeO_3	3 + 1	pyramidal

> **Solved example.** Why is H_3PO_3 dibasic?
> Its structure is $HP(O)(OH)_2$: one H is bonded directly to P and is not ionisable. Only the two O–H hydrogens are released.

Traps. Chlorine, not fluorine, has the most negative electron gain enthalpy. HF is the weakest hydrohalic acid but has the highest boiling point among HCl, HBr, HI and HF. Boric acid is a Lewis acid, not a protonic one. Count lone pairs on Xe from VSEPR: 8 valence electrons minus those used in bonds, in pairs.